Image Metadata Forensics

Python-Based Digital Forensic Tool for Advanced Validation & Analysis

The Need for Digital Truth

- ▹ **Legal Reliability:** Courts require irrefutable proof of origin and an unaltered state for digital imagery to be admissible.
- ▹ **Volume of Evidence:** Manual metadata extraction is impossible at scale; automated parsing is mission-critical for triage.
- ▹ **Uncovering Intent:** Hidden metadata often reveals the true timeline and geographical reality of a suspect's actions, contradicting their statements.

Multi-Format Extraction

JPEG & TIFF

Deep EXIF tag reading using **ExifRead** to bypass standard OS limitations and pull raw, unadulterated byte structures.

HEIC / HEIF

Native parsing of High-Efficiency formats common in modern iOS devices, extracting live-photo structures and dual-lens depth data.

XMP Parsing

Custom XML extraction to identify Adobe software edits and retrieve a granular image history beyond basic standard EXIF tags.

Precision GPS Decoding

Rational Translation

EXIF stores coordinates as rational tuples (DMS). We convert these to decimal format for mapping engine compatibility.

$$Dec = D + \frac{M}{60} + \frac{S}{3600}$$

Reference Validation

Coordinates are meaningless without polarity. We cross-reference **GPSLatitudeRef** and **GPSLongitudeRef** to apply correct negative vectors (South/West).

Altitude Precision

Extracting **GPSAltitude** tags to determine vertical positioning, crucial for modern multi-story drone or urban device investigations.

Interactive Trail Mapping

Visualizing the Digital Trail

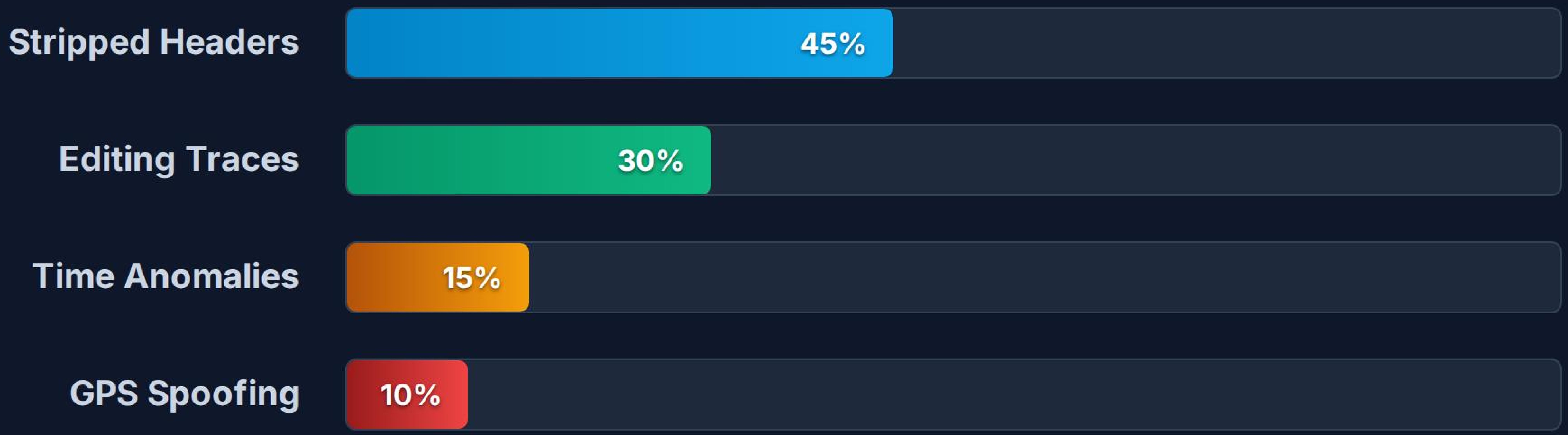
Data visualization bridges the critical gap between raw hexadecimal bytes and actionable intelligence.

- ▹ **Folium Integration:** Dynamic HTML maps generated purely from Python logic.
- ▹ **Chronological Routing:** Path lines drawn sequentially based on embedded DateTime tags.
- ▹ **Visual Corroboration:** Instantly map suspects' claimed locations against metadata reality.

Forensic Anomaly Alerts

- **Stripped Metadata:** Automatically flags images completely missing EXIF headers, highly indicative of intentional scrubbing or social media compression algorithms.
- **Manipulation Signatures:** Scans the 'Software' and XMP tags for editing suite fingerprints (e.g., Adobe Photoshop, GIMP) to immediately challenge image integrity.
- **Temporal Inconsistencies:** Cross-references *DateTimeOriginal* with *DateTimeDigitized*. Flags "future" timestamps or modified dates that precede creation dates.
- **Orphaned Location Data:** Detects cases where GPS blocks exist but corresponding timestamps are wiped, suggesting targeted metadata tampering.

Common Metadata Anomalies



Statistical frequency of metadata anomalies encountered in standard digital forensic ingestion pipelines. Our tool specifically targets these core discrepancies to accelerate the triage process.

Robust Chain of Custody

1. Ingest & Hash

Image bytes are immediately hashed using **SHA-256** to create an immutable fingerprint before parsing.

2. Evidence ID

Assignment of a globally unique UUID combined with the ingest timestamp for database tracking.

3. Sandboxed Parse

Metadata is read in a strictly *read-only* memory buffer, ensuring zero bytes are altered on disk.

4. Secure Output

Generation of tamper-proof JSON and PDF reports, signing extracted data alongside the original hash.

System Architecture

Python Core

Built on Python 3.11+, providing native cross-platform compatibility across Windows, macOS, and Linux forensic workstations.

Parsing Engine

Modular extraction blocks utilizing Pillow, ExifRead, and custom Regex, engineered to fail gracefully on corrupted byte streams.

Reporting Layer

Exports machine-readable JSON for automated SIEM ingestion and formatted PDFs for direct, unalterable submission to legal counsel.

Securing Digital Truth

Deploying Python to automate, validate, and secure the silent witness of image metadata in modern investigations.

Questions & Demonstrations

GitHub Repository Access Available Upon Request